Dear Hiring Manager,

I am writing to express my interest in the Data Scientist position at Orange Business. I recently completed my Master of Science in Computer Science with a specialization in Artificial Intelligence and Machine Learning from Sapienza University of Rome, where I developed a strong foundation in machine learning, natural language processing, computer vision, and AI system development.

Throughout my academic and project experience, I have focused on building and evaluating machine learning solutions while gaining hands-on experience with Python, PyTorch, FastAPI, Docker, and AWS. My master's thesis involved evaluating pretrained Vision Transformer architectures under continual learning settings, where I designed experimental pipelines, analyzed model robustness, investigated catastrophic forgetting, and benchmarked performance across sequential learning tasks. This work strengthened my skills in model evaluation, experimentation, and translating research findings into practical insights.

In addition to my research experience, I developed DocDelta, an AI-powered document intelligence platform that leverages transformer-based NLP models for semantic document comparison and analysis. As part of this project, I built scalable backend APIs using FastAPI, integrated machine learning models into production workflows, containerized the application using Docker, and deployed it on AWS ECS Fargate. This experience provided valuable exposure to the end-to-end lifecycle of AI applications, from model integration and inference pipelines to deployment and monitoring.

I have also contributed to machine learning research through an IEEE-published study on air pollution prediction, where I evaluated and compared multiple machine learning algorithms and analyzed their performance across environmental datasets. This experience enhanced my ability to conduct rigorous experimentation, apply statistical analysis, and communicate technical findings effectively.

What particularly attracts me to Orange Business is the opportunity to work on innovative AI and Large Language Model solutions within a global technology organization. The role's emphasis on machine learning research, model development, deployment, and cross-functional collaboration aligns closely with both my academic background and career aspirations. I am excited by the opportunity to contribute to building scalable AI systems while continuing to learn from experienced engineers and researchers.

I would welcome the opportunity to discuss how my background, technical skills, and enthusiasm for AI can contribute to Orange Business. Thank you for your time and consideration. I look forward to hearing from you.

Sincerely,

Swati Aggrawal